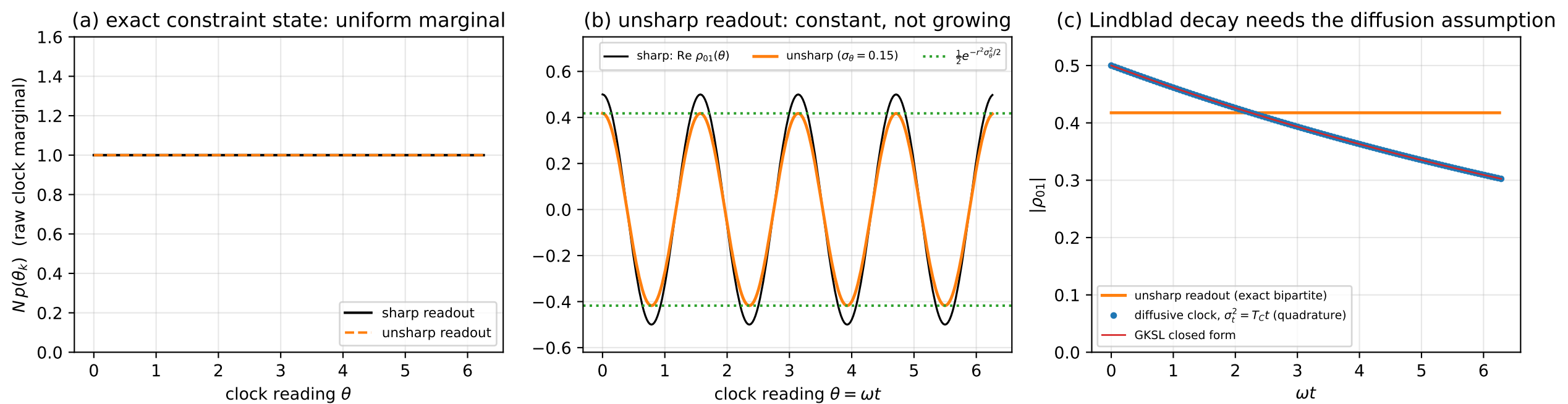
(a) exact constraint state: uniform marginal
$N\,p(\theta_k)$ (raw clock marginal)
clock reading θ
sharp readout
unsharp readout
(b) unsharp readout: constant, not growing
sharp: Re $\rho_{01}(\theta)$
unsharp ($\sigma_\theta = 0.15$)
$\frac{1}{2}e^{-r^2\sigma_\theta^2/2}$
clock reading $\theta = \omega t$
(c) Lindblad decay needs the diffusion assumption
$|\rho_{01}|$
ωt
unsharp readout (exact bipartite)
diffusive clock, $\sigma_t^2 = T_C t$ (quadrature)
GKSL closed form